

RESEARCH PROFILE

Interdisciplinary researcher working at the intersection of **psychology**, **computational psycholinguistics**, **cognitive science**, and **natural language processing**, with a focus on how AI systems affect human **belief**, **trust**, and **decision-making**. My research combines **behavioral experiments**, **psychological theory**, **context-aware design**, and **computational modeling** to identify where human and AI processing of social and psycholinguistic cues diverge, and what that divergence means for **trust**, **deception**, and **safety** in naturalistic, deployed systems.

Research focus:

- **Predictive processing** and **human–AI behavioral divergence and evaluation** – investigating how humans and artificial agents deal with the nuances of language (e.g., disfluency, alternative meanings) in multimodal settings
- **Trust, deception, and manipulation via psycholinguistic cues** – how features like hesitation markers or hedging language shape human trust and scrutiny of AI-generated content
- **Human-centered, context-aware intelligent systems** – improving interpretability and human–AI interaction quality
- **Emotion regulation** and **social-psychological responses to adversity** – focusing on behavioral, cognitive, and interpersonal strategies

Methods and Tools: Behavioral experimentation (eye-tracking, self-paced reading, cloze prediction); survey design; statistical modeling (R: regression, Bayesian, and multivariate methods, permutation-based analyses); language model evaluation pipelines (Python: PyTorch, Hugging Face Transformers, Pandas) comparing model outputs to human behavioral baselines, including cross-tokenization alignment and divergence metrics (e.g., Jensen-Shannon divergence); currently extending work to multimodal integration and agent-based modeling.

Applications: Conversational AI, embodied AI, human-AI trust and interaction quality, AI safety and societal risk evaluation, e-mental health technologies, human-centered system design, multimodal and cross-channel communication.

EDUCATION

Ph.D. Natural Language Processing *University of Edinburgh, UK* — 09/2021–06/2026

- Thesis: *Written Disfluencies and Nonliteral Meaning in Natural Language Processing: Evidence from Behavioral and Computational Studies*
- Supervisors: Martin Corley, Patrick Sturt

M.Sc. General Psychology *University of Tabriz, Iran* — 09/2018–04/2021

- Thesis: *Comparing the Aspects of Emotional Regulation in Individuals With and Without Sexual Traumatic Experiences*
- Supervisors: Mostafa Zarean, Darejan Jana Javakhishvili

M.A. English Language and Literature *Shahid Chamran University, Iran* — 09/2010–09/2013

- Thesis: *A Social Psychological Analysis of Power Distribution in José Saramago’s Blindness (Based on the 1997 English Translation by Giovanni Pontiero)*
- Supervisors: Mahmoud Daram, Saleh Hosseini

B.A. English Language and Literature *Shahid Chamran University, Iran* — 09/2006–07/2010

Took a career break between 2013 and 2018 for full-time caregiving reasons.

PUBLICATIONS (LONG PAPERS, SHORT PAPERS, BOOK CHAPTER, ABSTRACTS)

Tarighat, A., Sturt, P., Corley, M. (2025). It’s Umm Hard to Process but Not Hard to Evaluate: Written Disfluency Affects Humans and Language Models Differently. *The British Psychological Society – Scottish Branch*.

Tarighat, A., Sturt, P., Corley, M. (2024). Perspectives on Language Model and Human Handling of Written Disfluency and Nonliteral Meaning. *Proceedings of the 28th Workshop on the Semantics and Pragmatics of Dialogue (Trentologue)*.

Kapron-King, A., Fletcher, L. **Tarighat, A.**, Droop, S., Dobрева, R., Cummins, C., & Rohde, H. (2024) Tell it like it (usually) is(n't): Speakers’ mention of instruments in Reddit corpus reflects instrument (a)typicality. *Conference on Rational Approaches in Language Science (RAILS)*.

Hew, J., **Tarighat, A.**, Corley, M., Horne, Z. (2024). Examining structural and semantic predictors of announced sarcasm on r/AskReddit. *Proceedings of the Annual Meeting of the Cognitive Science Society*.

Tarighat, A., Corley, M. (2023) When Speech Becomes Writing: The Case of Disfluencies. *CLASP Conference on Learning with Small Data*.

Tarighat, F. S., Magdy, W., & Corley, M. (2022). Understanding Fillers May Facilitate Automatic Sarcasm Comprehension: A Structural Analysis of Twitter Data and a Participant Study. *Proceedings of the 26th Workshop on the Semantics and Pragmatics of Dialogue (DubDial)*.

Zarean, M., & **Tarighat, F. S.** (2022). Hearing the untold: a review of Central Asia’s contribution to the expansion of psychological assessment. *Comparative Histories of Psychological Assessment*. Cambridge University Press.

Tarighat, F. S. & Zarean, M. (2021). Cognitive Emotion Regulation in Individuals With and Without Sexual Traumatic Experiences. *National Congress of Family Psychology Association of Iran: The Family in Our Age*. University of Ahvaz. (Persian)

Zarean, M., **Tarighat, F. S.**, & Clough, B. (2019). Preferences and attitudes toward e-mental health services among Iranians. *The 1st Biennial Congress on Psychology and Behavioral Sciences*. University of Tabriz. (Persian)

Tarighat, F. S. (2019). A social psychological study of power relations considering neutral agents and gendered bodies in José Saramago’s *Blindness*. *The 1st Biennial Congress on Psychology and Behavioral Sciences*. University of Tabriz.

(In preparation)

Tarighat, A., Sturt, P., Corley, M. Umm, What Comes Next? Human and LM Predictions Under Written Filler.

Tarighat, A., Sturt, P., Corley, M. It’s Um Hard to Process but Not Hard to Evaluate: Written Fillers and Sarcasm in Reading and Evaluation Experiments. *Discourse Processes*.

Tarighat, A., Zarean, M., & Javakhishvili, J. Childhood Sexual Trauma Selectively Disrupts Cognitive and Interpersonal Emotion Regulation in Adults. *European Journal of Psychotraumatology*.

PROFESSIONAL EXPERIENCES (RESEARCH, SUPERVISION, TEACHING, TRANSLATION, AND MANAGEMENT)

- Ph.D. Graduate Assistant, psycholinguistics project supervision of undergraduate students, Department of Psychology, University of Edinburgh (2023–2026)
- Research Collaborator, with Julian Hough, Swansea University (2025–Present)
 - Examining the universality of disfluencies using the DUEL multilingual corpus
 - Devising labeling and evaluation tasks across languages with insights from neural and incremental mechanisms of disfluency detection
- M.Sc. Graduate Assistant, standardized psychology questionnaire translation and tailoring for graduate research projects, Department of Psychology, University of Tabriz (2018–2021)
- Freelance English Language Instructor, In-Person/Online Courses (2006-2021)
- Senior Language Instructor (2006–2018)
- Freelance Scientific and Literary Translator and Editor (2008–present)
- International Affairs Manager, Mashal Tarigh Tejarat Aria Trading Company, Urmia, Iran (2009–2010)

TRAINING, SKILLS, AND INTERESTS

Programming, Analysis, and Research

- **Python** (Hugging Face, Pandas, NumPy, PyTorch, spaCy)
- **R** (data wrangling, regression, Bayesian modeling)
- **JavaScript** (experiment design)
- **Experimentation:** Behavioral experiments (eye-tracking and self-paced reading, ExperimentBuilder, DataViewer, E-Prime, OpenSesame); Survey design (Qualtrics); Standardized questionnaires and structured interviews

Academic and Professional Training

- **Selected courses (Ph.D.):** Univariate & Multivariate Statistical Modeling in R; Computer Programming for Speech and Language Processing; Pragmatics; Doing Research in NLP; Accelerated NLP; Case Studies in AI Ethics; Text Technologies for Data Science; Machine Learning Practical
- **Other:** Interior architecture and design (certified)
- **Languages:** English, Persian, Azerbaijani, Turkish (+ basic French, German, Suret)

Additional

- **Trauma care psychology:** Corrective exercises for posttraumatic stress care

AWARDS AND SCHOLARSHIPS

- UKRI Centre for Doctoral Training in Natural Language Processing Scholarship, University of Edinburgh, UK, 2021–2025
- Exceptional Talents Award for Higher Education Programs Enrollment, Shahid Chamran University, Iran, 2010
- Undergraduate Scholarship for Shahid Chamran University Obtained via the Annual National University Entrance Exam, Iran, 2006–2010

INVITED TALKS

- CDTalks, Student-led Seminars by CDT in NLP, *Language Model and Human Handling of Written Disfluency and Nonliteral Meaning*, Edinburgh, 2024
- Brain, Cognition, and Development Seminar *Exploring Structural and Semantic Predictors of Written Sarcasm in Online Settings*, Edinburgh, 2024
- CDTalks, Student-led Seminars by CDT in NLP, *Differences in Human and Machine Interpretation of Nonliteral Meanings*, Edinburgh, 2023
- Psycholinguistics Coffee, Seminars at the School of PPLS, *When Speech Becomes Writing: The Case of Disfluencies*, Edinburgh, 2023
- Alan Turing Institute, Student Networking and Presentation Session, *Tracing Sarcastic Meanings of Fillers on Twitter: A Structural Analysis of Data and a Pilot Study*, London, 2022

WORKSHOPS AND CONFERENCES (SPEAKER, PRESENTER, TRAINEE, ATTENDEE)

- The British Psychological Society – Scottish Branch, Perth (2025) - Speaker
- UKRI AI CDTs in NLP Joint Partners’ Summit, Edinburgh (2025) - Poster presenter
- The 28th Workshop on the Semantics and Pragmatics of Dialogue, SemDial (TrentoLogue), Rovereto, Italy (2024) - Speaker
- The CLASP Conference on Learning with Small Data (LSD), Gothenburg, Sweden (2023) - Poster presenter
- The 26th Workshop on the Semantics and Pragmatics of Dialogue, SemDial (DubDial), Dublin, Ireland (2022) - Poster presenter
- Machine Learning in Cognitive Sciences, University of Tabriz, Tabriz, Iran (2021) - Trainee
- Cognitive Rehabilitation: Attention, Memory, and Executive Functions, Shenakht Center, Tehran, Iran (2020) - Trainee
- National Congress of Family Psychology Association of Iran: The Family in Our Age, Ahvaz, Iran (2021) - Speaker
- The 1st Biennial Congress on Psychology and Behavioral Sciences, University of Tabriz, Tabriz, Iran (2019) - Attendee
- Introduction to EEG Biofeedback (Neurofeedback), Approved by the Biofeedback Federation of Europe (BFE), Urmia, Iran (2019) - Trainee
- Premarital Counseling, Iranian Clinical Psychology Association, Tabriz, Iran (2018) - Trainee
- Transdiagnostic Treatments for Anxiety Disorders, Clinical Psychology Association, Urmia, Iran (2018) - Trainee
- The 2nd National Conference on Literary Criticism, Tarbiat Modares University, Tehran, Iran (2011) - Attendee

VOLUNTEERING AND COMMUNITY SERVICE

- Recruiting Underrepresented Gifted Students for Educational Funds, SEKHAVAT Coordination Center for the Benefactors of Urmia, Urmia, Iran (2018–2021)
- Cancerous Patients’ Emotional Support, Omid Hospital, Urmia, Iran (2013–2015)
- Language Instructor, General English Program for Cancerous Students, Shafa Hospital, Ahvaz, Iran (2007)

- Student Assistant, Psychology Support Team for Runaway Girls with a Focus on Sexual Abuse and Assault Survivors (2006–2007)

COMMUNITY ENGAGEMENT

- **Co-organizer**, Interdisciplinary Writing Retreat for Doctoral Researchers, University of Edinburgh, 2024
Designed and co-led a 4-day academic writing retreat to Northumberland for supporting CDT in NLP cohorts.
- **Co-organizer**, Nature Retreats at Fimbush, 2022–2023
Helped organize 3-day retreats integrating nature, informal dialogue, scientific problem-solving, and reflective practice for CDT in NLP researchers.
- **Organizer**, Informatics Postgraduate Walk + Picnic, University of Edinburgh, 2023
Facilitated a casual community-building event fostering peer support and wellbeing among early-career Informatics researchers.
- **Volunteer**, Architecture and Mechanisms for Language Processing (AMLaP), 2024
Supported conference logistics, participant coordination, and venue setup as part of the organizing team.

REFERENCES

- **Prof. Martin Corley**, School of PPLS, University of Edinburgh. Martin.Corley@ed.ac.uk
- **Prof. Hannah Rohde**, School of PPLS, University of Edinburgh. Hannah.Rohde@ed.ac.uk
- **Dr. Patrick Sturt**, School of PPLS, University of Edinburgh. Patrick.Sturt@ed.ac.uk
- **Prof. Daniel Mirman**, School of PPLS, University of Edinburgh. daniel.mirman@ed.ac.uk
- **Dr. Mostafa Zarean**, Department of Psychology, University of Tabriz. zarean@tabrizu.ac.ir